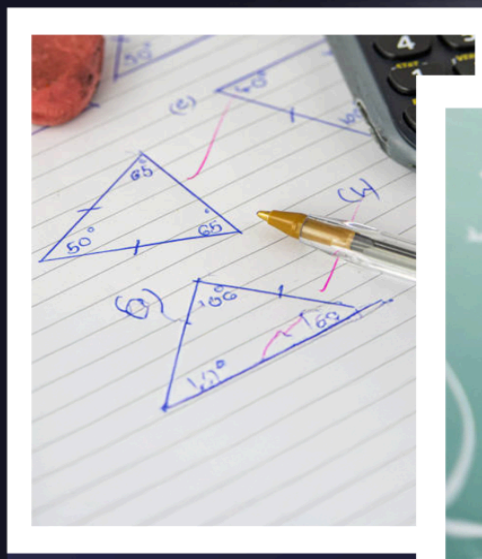
testbook PASS PRO MAX

Quantitative Aptitude

Formula – Guide

Trigonometry



Trigonometric

Trigonometric ratios

$$\sin\theta = \frac{\text{Perpendicular}}{\text{Hypotenuse}}$$

$$\cos\theta = \frac{\text{Base}}{\text{Hypotenuse}}$$

$$\tan\theta = \frac{\text{Perpendicular}}{\text{Base}}$$

$$\csc\theta = \frac{\text{Hypotenuse}}{\text{Perpendicular}}$$

$$\sec\theta = \frac{\text{Hypotenuse}}{\text{Base}}$$

$$\cot\theta = \frac{\text{Base}}{\text{Perpendicular}}$$

Values of Trigonometric ratios:

Angle (Degrees)	0°	30°	45°	60°	90°
$\sin\theta$	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1
$\cos\theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0

$\tan\theta$	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	Not Defined
$\csc\theta$	Not Defined	2	$\sqrt{2}$	$\frac{2}{\sqrt{3}}$	1
$\sec\theta$	1	$\frac{2}{\sqrt{3}}$	$\sqrt{2}$	2	Not Defined
$\cot\theta$	Not Defined	$\sqrt{3}$	1	$\frac{1}{\sqrt{3}}$	0

Ratios of $-\theta$ in Terms of those of θ :

1. $\sin(-\theta) = -\sin\theta$
2. $\cos(-\theta) = \cos\theta$
3. $\tan(-\theta) = -\tan\theta$
4. $\text{cosec}(-\theta) = -\text{cosec}\theta$
5. $\sec(-\theta) = \sec\theta$
6. $\cot(-\theta) = -\cot\theta$

Ratios of $90 - \theta$ in Terms of those of θ :

1. $\sin(90^\circ - \theta) = \cos\theta$
2. $\cos(90^\circ - \theta) = \sin\theta$
3. $\tan(90^\circ - \theta) = \cot\theta$
4. $\text{cosec}(90^\circ - \theta) = \sec\theta$
5. $\sec(90^\circ - \theta) = \text{cosec}\theta$
6. $\cot(90^\circ - \theta) = \tan\theta$

Ratios of $180 - \theta$ in Terms of those of θ:

1. $\sin(180° - \theta) = \sin\theta$
2. $\cos(180° - \theta) = -\cos\theta$
3. $\tan(180° - \theta) = -\tan\theta$
4. $\text{cosec}(180° - \theta) = \text{cosec}\theta$
5. $\sec(180° - \theta) = -\sec\theta$
6. $\cot(180° - \theta) = -\cot\theta$

Ratios of $270 - \theta$ in Terms of those of θ:

1. $\sin(270° - \theta) = -\cos\theta$
2. $\cos(270° - \theta) = -\sin\theta$
3. $\tan(270° - \theta) = \cot\theta$
4. $\text{cosec}(270° - \theta) = -\sec\theta$
5. $\sec(270° - \theta) = -\text{cosec}\theta$
6. $\cot(270° - \theta) = -\tan\theta$

Ratios of $360 - \theta$ in Terms of those of θ:

1. $\sin(360° - \theta) = -\sin\theta$
2. $\cos(360° - \theta) = \cos\theta$
3. $\tan(360° - \theta) = -\tan\theta$
4. $\text{cosec}(360° - \theta) = -\text{cosec}\theta$
5. $\sec(360° - \theta) = \sec\theta$
6. $\cot(360° - \theta) = -\cot\theta$

Ratios of $90 + \theta$ in Terms of those of θ:

1. $\sin(90° + \theta) = +\cos\theta$
2. $\cos(90° + \theta) = -\sin\theta$

3. $\tan(90° + \theta) = -\cot\theta$
4. $\text{cosec}(90° + \theta) = +\sec\theta$
5. $\sec(90° + \theta) = -\text{cosec}\theta$
6. $\cot(90° + \theta) = -\tan\theta$

Ratios of $180 + \theta$ in Terms of those of θ:

1. $\sin(180° + \theta) = -\sin\theta$
2. $\cos(180° + \theta) = -\cos\theta$
3. $\tan(180° + \theta) = +\tan\theta$
4. $\text{cosec}(180° + \theta) = -\text{cosec}\theta$
5. $\sec(180° + \theta) = -\sec\theta$
6. $\cot(180° + \theta) = +\cot\theta$

Ratios of $270 + \theta$ in Terms of those of θ:

1. $\sin(270° + \theta) = -\cos\theta$
2. $\cos(270° + \theta) = +\sin\theta$
3. $\tan(270° + \theta) = +\cot\theta$
4. $\text{cosec}(270° + \theta) = -\sec\theta$
5. $\sec(270° + \theta) = +\text{cosec}\theta$
6. $\cot(270° + \theta) = -\tan\theta$

Ratios of $360 + \theta$ in Terms of those of θ:

1. $\sin(360° + \theta) = +\sin\theta$
2. $\cos(360° + \theta) = +\cos\theta$
3. $\tan(360° + \theta) = +\tan\theta$
4. $\text{cosec}(360° + \theta) = +\text{cosec}\theta$
5. $\sec(360° + \theta) = +\sec\theta$

6. $\cot(360^\circ + \theta) = +\cot\theta$

Some Important trigonometric values

- $\sin 18^\circ = \frac{\sqrt{5}-1}{4}$
- $\sin 150^\circ = \frac{\sqrt{3}-1}{2\sqrt{2}} = \cos 75^\circ$
- $\cos 18^\circ = \frac{\sqrt{10+2\sqrt{5}}}{4}$
- $\cos 15^\circ = \frac{\sqrt{3}+1}{2\sqrt{2}} = \sin 75^\circ$
- $\cos 36^\circ = \frac{\sqrt{5}+1}{4}$
- $\sin 22.5^\circ = \frac{\sqrt{2-\sqrt{2}}}{2}$
- $\sin 36^\circ = \frac{\sqrt{10-2\sqrt{5}}}{4}$
- $\cos 22.5^\circ = \frac{\sqrt{2+\sqrt{2}}}{2}$

Useful TriAgonometric Results:

If $A + B = 90^\circ$, then:

- $\tan A \cdot \tan B = 1$
- $\cot A \cdot \cot B = 1$
- $\sin A \cdot \sec B = 1$
- $\cos A \cdot \operatorname{cosec} B = 1$
- $\sin^2 A + \sin^2 B = 1$
- $\cos^2 A + \cos^2 B = 1$
- $\sin A = \cos B$

- $\operatorname{cosec} A = \sec B$
- $\tan A = \cot B$

Trigonometric Identities

- $cosec\theta = \frac{1}{sin\theta}$
- $cosec\theta = \frac{1}{sin\theta}$
- $cosec\theta = \frac{1}{sin\theta}$
- $cosec\theta = \frac{1}{sin\theta}$
- $cosec\theta = \frac{1}{sin\theta}$
- $sec^2\theta - tan^2\theta = 1$
- $(sec\theta - tan\theta)(sec\theta + tan\theta) = 1$
- **If** $sec\theta \pm tan\theta = x$ **than,** $se\,c\theta \mp tan\theta = \frac{1}{X}$
- $cosec^2\theta - cot^2\theta = 1$
- $(cosec\theta - cot\theta)(cosec\theta + cot\theta) = 1$

Relationship between Trigonometry ratios

- $sin^2\theta + cos^2\theta = 1$
- $sin^2\theta = 1 - cos^2\theta$
- $cos^2\theta = 1 - sin^2\theta$
- $sec^2\theta = 1 + tan^2\theta$
- $cosec^2\theta = 1 + cot^2\theta$

Sum And Difference Formulas

- $sin(A + B) = sinA \cdot cosB + cosA \cdot sinB$

- $sin(A - B) = sinA \cdot cosB - cosA \cdot sinB$
- $cos(A + B) = cosA \cdot cosB - sinA \cdot sinB$
- $cos(A - B) = cosA \cdot cosB + sinA \cdot sinB$
- $2 \cdot sinA.cosB = sin(A + B) + sin(A - B)$
- $2 \cdot cosA.sinB = sin(A + B) - sin(A - B)$
- $2 \cdot sinA.sinB = cos(A - B) - cos(A + B)$
- $2.cosA.cosB = cos(A + B) + cos(A - B)$
- $sin^2A - sin^2B = sin(A + B).sin(A - B)$
- $cos^2A - cos^2B = cos(A + B).cos(A - B)$

Half Angle Formula

- $sinC + sinD = 2sin\left(\frac{C+D}{2}\right) \cdot cos\left(\frac{C-D}{2}\right)$
- $sinC - sinD = 2cos\left(\frac{C+D}{2}\right) \cdot sin\left(\frac{C-D}{2}\right)$
- $cosC + cosD = 2cos\left(\frac{C+D}{2}\right) \cdot cos\left(\frac{C-D}{2}\right)$
- $cosC - cosD =- 2sin\left(\frac{C+D}{2}\right) \cdot sin\left(\frac{C-D}{2}\right)$

Tangent Formulas

- $tan(A + B) = \frac{tanA+tanB}{1-tanA \cdot tanB}$
- $tan(A - B) = \frac{tanA-tanB}{1+tanA \cdot tanB}$
- $cot(A + B) = \frac{cotA+cotB}{cotA \cdot cotB-1}$
- $cot(A - B) = \frac{cotB-cotA}{cotA \cdot cotB+1}$

Identities for Multiple Angles

- $sin2A = 2sinA \cdot cosA = \frac{2tanA}{1+tan^2A}$
- $cos2A = cos^2A - sin^2A = 2cos^2A - 1 = 1 - 2sin^2A = \frac{1-tan^2A}{1+tan^2A}$
- $tan2A = \frac{2tanA}{1-tan^2A}$
- $sin3A = 3sinA - 4sin^3A$
- $cos3A = 4cos^3A - 3cosA$
- $tan3A = \frac{3tanA-tan^3A}{1-3tan^2A}$

Maximum and Minimum Values

1. **Sine Function:** $-1 \leq sin\theta \leq 1$
2. **Cosine Function:** $-1 \leq cos\theta \leq 1$
3. **Tangent Function:** $-\infty < tan\theta < \infty$
4. **Cotangent Function:** $-\infty < cot\theta < \infty$
5. **Secant Function:** $sec\theta \geq 1$ or $sec\theta \leq -1$
6. **Cosecant Function:** $csc\theta \leq -1$ or $csc\theta \geq 1$

Limits and Properties of Sine and Cosine Functions:

1. **Limits of Sine Function:** $0 \leq sin^2\theta \leq 1$
2. **Limits of Cosine Function:** $0 \leq cos^2\theta \leq 1$
3. **Inequalities for Sine:** $sin^2\theta \leq |sin\theta| \leq 1$

4. **Inequalities for Cosine:** $cos^2\theta \leq |cos\theta| \leq 1$

Maximum and Minimum Values of $asin\theta \pm bcos\theta$

- Maximum Value: $+\sqrt{a^2 + b^2}$
- Minimum Value: $-\sqrt{a^2 + b^2}$

Maximum and Minimum values of $p = asin^2\theta + bcos^2\theta$

- If $a < b$ then Maximum = b & Minimum = a
- If $a > b$ then Maximum = a & Minimum = b

Maximum and Minimum Values of $p = atan^2\theta + bcot^2\theta$

If $a > 0 \& b > 0$ **then** $Maximum = \infty \,\&\, Minimum = 2\sqrt{ab}$

Maximum and Minimum Values of

$p = asin^2\theta + bcsc^2\theta$

$p = acos^2\theta + bsec^2\theta$

$p = atan^2\theta + bcot^2\theta$

If $a > 0 \,\&\, b > 0$ then,

- $Maximum = \infty \,\&\, Minimum = 2\sqrt{ab}\ when\ (a \geq b)$
- $Maximum = \infty \,\&\, Minimum = (a + b)\ when\ (a \leq b)$

Minimum Value of $sin^n\theta \cdot cos^n\theta$

- If n is odd: $Min\left(sin^n\theta \cdot cos^n\theta\right) = \left(-\frac{1}{2}\right)^n$
- If n is even: $Min\left(sin^n\theta \cdot cos^n\theta\right) = 0$

Maximum Value of $sin^n\theta \cdot cos^n\theta$

- $Max\left(sin^n\theta \cdot cos^n\theta\right) = \left(\frac{1}{2}\right)^n$

Maximum Value of $sin^n\theta + cos^m\theta$

Condition: For m, n ≥ 2 and Interval: 0 ≤ 90

- $max\left(sin^n\theta + cos^m\theta\right) = 1$

Minimum Value of $asec^2\theta + bcsc^2\theta$

- $asec^2\theta + bcsc^2\theta = (\sqrt{a} + \sqrt{b})^2$

Conversion Between Degrees and Radians

From degrees to radians:

- $radians = \left(\frac{\pi}{180}\right) \times degrees$

From radians to degrees:

- $degrees = \left(\frac{180}{\pi}\right) \times radians$

Special Angles in Radians:

Degrees	Radians
0°	0
30°	$\frac{\pi}{6}$
45°	$\frac{\pi}{4}$
60°	$\frac{\pi}{3}$

90°	$\frac{\pi}{2}$
180°	π
270°	$\frac{3\pi}{2}$
360°	2π

- $Arc\ length\ = r \times \theta$
- $Area\ of\ sector\ = \frac{1}{2} r^2 \theta$

Degree, Minute, and Second

- 1 degree $(^\circ) = \frac{1}{90}$ of a right angle.
- 1 minute $(1^{'}) = \frac{1}{60}$ of a degree.
- 1 second $(1^{''}) = \frac{1}{60}$ of a minute.

Additional info:

- $90^\circ = 1$ right angle.
- $60^{'} = 1^\circ$
- $60^{''} = 1^{'}$